

THE MAIN COMPONENTS OF HARMONY OS: MICROKERNEL SECURITY SUPPORT



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PREFACE

In the name of Allah SWT. Praise and gratitude for His presence, mercy, and guidance so that the author can complete this writing task well and smoothly. This paper has been compiled by the author to the best of his ability and with the assistance of several parties. Therefore, the author would like to express his gratitude to those who contributed to the completion of this writing assignment. The author realizes that there are still many shortcomings and errors in this writing assignment. Therefore, we are open to any criticism and suggestions from readers so that the author can improve the writing.

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CHAPTER I INTRODUCTION

1.1 Background

In today's world, technological competitiveness in industry is increasing with the times. Being in Industry 5.0 requires humans to quickly adapt to machines or technology. This also affects increasingly sophisticated operating systems. It is important for society as a whole to learn about this and not be left behind. In addition, with increasingly sophisticated operating systems, this is also supported by security that protects data systems on the Internet. Therefore, this text focuses on the security components of the Harmony Operating System (HarmonyOS), introducing the security components used by Harmony OS. HarmonyOS is responsible for controlling and managing hardware commonly known under the brand name 'Huawei'. Recently, the Huawei brand has experienced growth and is now competing with iOS, which previously dominated the industry. This has led to HarmonyOS appearing to be superior to other operating systems. The next chapter will discuss HarmonyOS in depth, including its security components, advantages and disadvantages, and issues related to system security.

1.2 Writing Objective

The purpose of this research is to introduce the security components of HarmonyOS and how the system works. This includes security and privacy, star shields, microkernels, and other features. The main focus will be on how the security components work in HarmonyOS.

1.3 Problem Domain

This study examines how star shield and microkernel work in HarmonyOS. One of the main innovations in HarmonyOS is the use of microkernel architecture. This differs from the Android macrokernel operating system, which commonly uses the Linux kernel. The purpose of this study is to identify the significant advantages of using one of its main innovations, namely Microkernel.

1.4 Writing Methodology

This paper is based on articles, journals, and accurate qualitative and quantitative data. From the sources obtained, the next step is to describe, prove, and develop the information focusing

on the microkernel and star shield systems used to understand and explain to readers. This approach aims to create a systematic, factual, and accurate description that highlights the security system working on HarmonyOS and its advantages and disadvantages compared to other operating systems, thereby raising several issues that need to be evaluated and improved in the future.

1.5 Writing Framework

Chapter I: Introduction

HarmonyOS and the security components used by HarmonyOS to provide readers with an overview.

Chapter II: Basic Theory

Introduce readers to the basic material about HarmonyOS, which will be explained further in the following chapters. It will cover the operating systems of security components in HarmonyOS, such as Star Shield and Microkernel.

Chapter III: Problem Analysis

Problems in the HarmonyOS operating system and its security components.

Chapter IV: Conclusion and Suggestion

Conclusions about HarmonyOS, Star Shield, and Microkernel, as well as recommendations related to HarmonyOS.

CHAPTER II BASIC THEORY

2.1 What is an Operating System?

An operating system (OS) is a foundational software layer that manages a computer's hardware and software resources. It acts as a bridge between the physical hardware and the applications or end users, ensuring that hardware components are used effectively to perform various computing tasks. By providing essential services, the OS enables applications to run and interact with hardware without needing to manage hardware directly.

2.2 What is HarmonyOS?

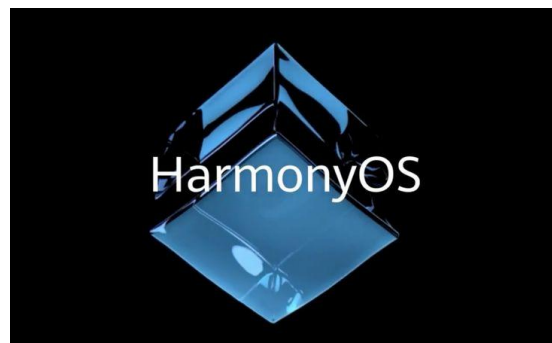


Figure 2.2 HarmoyOS

(Ref: <https://id.pinterest.com/pin/659847782886878422/>)

abbreviated as HMOS, is an innovative operating system developed by Huawei, a leading Chinese technology company. First introduced in 2019, HarmonyOS is designed to provide better integration between devices in the Internet of Everything (IoE) ecosystem. Unlike traditional operating systems such as Android and Windows, which function independently on specific devices, HarmonyOS has the flexibility to operate on various types of devices. These include smartphones, smart TVs, smart watches (wearables), and smart home devices, enabling all of these devices to communicate with each other and share resources more efficiently.

2.3 How the history of HarmonyOS?

The history of HarmonyOS, originally known as Hongmeng, began on August 9, 2019, when Huawei officially launched this operating system at its first developer conference in Dongguan. At the launch, Huawei described HarmonyOS as a microkernel-based distributed operating system designed for various types of hardware. This Chinese-developed operating system (OS) completely discards the Linux kernel and Android Open Source Project (AOSP) code base and supports a fully independent architecture built around Huawei's proprietary Hongmeng kernel and system. With a primary focus on IoT (Internet of Things) devices, including smart TVs, wearables, and in-car entertainment systems, Huawei does not explicitly position HarmonyOS as a mobile OS. This approach demonstrates Huawei's commitment to creating a more integrated and flexible ecosystem.

Following its initial launch, Huawei unveiled HarmonyOS 2.0 on September 10, 2020, at the Huawei Developer Conference. In the announcement, the company stated its plans to implement the operating system into its smartphones in 2021. The first developer beta for HarmonyOS 2.0 was released on December 16, 2020, along with DevEco Studio and a cloud emulator that allows developers to start adapting to this new platform.

On June 2, 2021, Huawei officially launched HarmonyOS 2.0 and released new devices equipped with this operating system. In addition, the company began rolling out updates for older devices in stages, allowing more users to enjoy the benefits of this new operating system. With each new version, HarmonyOS continues to evolve, with improvements in stability and performance, as well as compatibility with the latest hardware.

2.4 What Is Security System?

A security system is a series of devices and procedures designed to protect individuals, property, and information from threats such as theft, fire, sabotage, and other criminal activities. These systems can be physical, such as locks and fences, or digital, such as firewalls and antivirus software. The core of a security system is to effectively detect, prevent, and respond to threats, both in the home and business environments.

CHAPTER III PROBLEM ANALYSIS

3.1 How is HarmonyOS secured?

The main innovation in HarmonyOS is the use of a microkernel architecture. This differs from the macrokernel Android operating system, which commonly uses the Linux kernel. The microkernel architecture offers better performance and higher security compared to traditional operating systems. Microkernel architecture is a design approach in which the core functionality of a system is kept minimal and lightweight. The idea is to have a small, efficient kernel that only handles the most fundamental tasks, such as communication between components and basic system management. Additional features and services, like file systems, network protocols, or user interfaces, are implemented as separate modules or plugins outside the core kernel. These modules interact with the kernel through well-defined interfaces. This separation allows the system to be more flexible, modular, and easier to update or extend, as new features can be added without modifying the core system. This architecture is commonly used in operating systems and complex applications that require high adaptability and modularity.

3.2 Microkernel Architecture

Since the kernel is the core part of the operating system, so it is meant for handling the most important services only. Thus in this architecture, only the most important services are inside the kernel and the rest of the OS services are present inside the system application program. Thus users are able to interact with those not so important services within the system application. And the microkernel is solely responsible for the most important services of the operating system they are named as follows:

1. Interprocess Communication (IPC)

interprocess communication (IPC) is the sharing of data between running processes in a computer system, or between multiple such systems. Mechanisms for IPC may be provided by an operating system. Applications which use IPC are often categorized as clients and servers, where the client requests data and the server responds to client

requests. Many applications are both clients and servers, as commonly seen in distributed computing.

2. Memory Management

Memory Management is a critical aspect of operating systems that ensures efficient use of the computers memory resources. It controls how memory is allocated and deallocated to processes, which is key to both performance and stability. Below is detailed overview of the various components and techniques involved in memory management.

3. CPU scheduling

CPU scheduling is the process of temporarily controlling the CPU while other processes are suspended. The primary function of CPU scheduling is to ensure that whenever the CPU is idle, the operating system has selected at least one available process in the ready queue. In multiprogramming, if the long-term scheduler selects multiple I/O-bound processes, the CPU will be idle most of the time. The purpose of effective programming is to improve resource utilization.

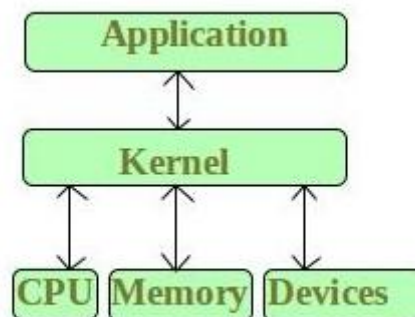


Figure 3.2 Microkernel Architecture

(Ref: <https://www.geeksforgeeks.org/operating-systems/microkernel-in-operating-systems/>)

3.3 Benefits of Microkernel Architecture

1. **High Security:** Microkernels operate by separating the core functions of the operating system from other functions, which helps reduce the risk of cyber-attacks and vulnerabilities. This makes the system more secure and reliable.

2. **Good Reliability:** Microkernel designs tend to be more stable, as unnecessary functions do not affect the core operations of the system. If one function fails, the system as a whole can still function properly.
3. **High Scalability:** With the ability to adapt to various devices, microkernels make HarmonyOS suitable for use on devices of various sizes, from small to large.
4. **Easy Maintenance:** The simpler structure of microkernels makes system maintenance and updates easier. Development teams can more quickly fix bugs or add new features without disrupting other functions.
5. **Support for Distributed Computing:** Microkernels enable distributed computing, where various devices can work together and share resources efficiently.

3.4 Microkernel-Based Operating System

Microkernel architectures are popular for specialized and mainstream uses. Examples include:

1. IBM AIX, a version of UNIX from IBM
2. Amoeba, a kernel developed for educational purposes
3. Mach kernel, used in the GNU/Hurd, NextSTEP, OPENSTEP, and Mac OS/X operating systems
4. Minix, a kernel developed by Andrew Tanenbaum for educational purposes
5. Symbian OS, an operating system popularly used on mobile phones, handheld devices, embedded devices, and PDA phones.

disadvantages of microkernel systems

1. Hardware must work through drivers.
2. Hardware may react more slowly because drivers are in user mode.
3. Processes must wait in a queue to obtain information.
4. Processes cannot access other processes without waiting.

CHAPTER IV CONCLUSION AND SUGGESTION

4.1 Conclusion

In conclusion, Huawei describes HarmonyOS as a microkernel-based distributed operating system designed for various types of hardware. Since the kernel is the core part of the operating system, it is designed to handle only the most important services. Therefore, in this architecture, only the most important services are located in the kernel, while other operating system services are located in the system application program.

4.2 Suggestion

Overall, microkernels perform well on the HarmonyOS operating system. Future improvements will further enhance the system's performance by reducing several issues that arise during use, including the shortcomings of microkernels.

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